6 ~~(a)~~

We apply Gram-Schmidt to build an orthonormal basis for M_k, that is $M_k = \text{span}\{v^1, .., v^k\}$ with

$$\langle v^i, v^j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}.$$

We then know that

$$\hat{x}_k = \hat{\alpha}_1 v^1 + \cdots + \hat{\alpha}_k v^k$$

where $\hat{\alpha}_i = \langle x, v^i \rangle$.

(a). Because $y_{k+1} \perp M_k = \text{span}\{v^1, .., v^k\}$, we define

$$v^{k+1} = \frac{y_{k+1}}{\|y_{k+1}\|}$$

k+1, sorry!

Hence, we have

$$\hat{x}_{k+1} = \hat{\alpha}_1 v^1 + \cdots + \hat{\alpha}_k v^k + \hat{\alpha}_{k+1} v^{k+1}$$

where $\hat{\alpha}_i = \langle x, v^i \rangle$.

Therefore,

$$\hat{x}_{k+1} = \hat{x}_k + \hat{\alpha}_{k+1} v^{k+1}$$

$$= \hat{x}_k + \alpha_{k+1} \frac{y_{k+1}}{\|y_{k+1}\|}$$

$\therefore$ $$\hat{x}_{k+1} = \hat{x}_k + \beta y_{k+1}$$

when $$\beta = \frac{1}{\|y_{k+1}\|} \langle x, v^{k+1} \rangle$$

$$= \frac{1}{\|y_{k+1}\|^2} \langle x, y_{k+1} \rangle$$

$$= \frac{\langle x, y_{k+1} \rangle}{\langle y_{k+1}, y_{k+1} \rangle}.$$

(b) From the Projection Theorem,

$$\cancel{\ } \; y_{k+1} - \hat{y}_{k+1|k} \perp M_k$$

moreover, $\hat{y}_{k+1|k} \in M_k$, and

thus $M_{k+1} = \text{span}\{y_1, \ldots, y_k, y_{k+1}\}$

$$= \text{span}\{y_1, \ldots, y_k, y_{k+1} - \hat{y}_{k+1|k}\}.$$

$$= \text{span}\{y_1, \ldots, y_k, \tilde{y}_{k+1}\}$$

where $\tilde{y}_{k+1} = y_{k+1} - \hat{y}_{k+1|k}$ is orthogonal to M_k. Hence, from (a)

$$\hat{x}_{k+1} = \hat{x}_k + \beta \tilde{y}_{k+1}$$

where $\beta = \dfrac{\langle x, \tilde{y}_{k+1} \rangle}{\langle \tilde{y}_{k+1}, \tilde{y}_{k+1} \rangle}$.

Substituting in for $\tilde{y}_{k+1}$, we have

$$\hat{x}_{k+1} = \hat{x}_k + \beta\left(y_{k+1} - \hat{y}_{k+1|k}\right)$$

where

$$\beta = \frac{\langle x, (y_{k+1} - \hat{y}_{k+1|k}) \rangle}{\langle y_{k+1} - \hat{y}_{k+1|k}, y_{k+1} - \hat{y}_{k+1|k} \rangle}$$

$\square$

Remark: There are may ways to work this problem. Your solution does not need to match this one.